TALKING WITH MATH:
WRITING & SOLVING EXPRESSIONS!

Turn Everyday Words into Magic Math Codes

Grade 4 Algebra Mastery Guide

Did you know math is its own secret language?

Let's learn how to translate normal story clues into powerful math sentences!

What is a Math "Expression"?

Think about the words you use when you talk. You combine words together to make a sentence phrase, like "three happy puppies."

In algebra, a **Math Expression** is exactly like a word phrase, but written using numbers, operation signs (like plus or minus), and letters instead of words!

Here are some examples of math expressions:

- $x + 4$ (A mystery number plus four)
- $10 - y$ (Ten minus a secret value)
- $3 \times n$ (Three times an unknown amount)

The Golden Secret:
An expression is a math phrase that does **NOT** have an equal sign (=) yet! It's just a grouping of items waiting to be calculated.

The Missing Equal Sign!

A lot of people get confused between an *expression* and an *equation*. Let's learn how to tell them apart like total math geniuses!

Think of it like a train track:

The Expression (The Starting Train)
It's just a phrase. No balance scale is built yet.

$$x + 5$$

The Equation (The Complete Station)
An equation uses a giant equal sign (=) to connect two parts together.

$$x + 5 = 12$$

Today, we are going to master writing the **expressions** first, so we can build any equation we want later!

Word Clues for Plus and Minus

To translate human stories into math codes, you have to look for special spy words!

Let's check out the secret vocabulary codebook for addition and subtraction:

ADDITION (Plus) Clues	SUBTRACTION (Minus) Clues
• More than • Sum of • Increased by • Added to	• Less than • Difference of • Decreased by • Taken away from

If a story says "5 more than a number", your brain should instantly scream: $n + 5$!

Word Clues for Times and Share

What about when things are growing bigger by groups, or split up into fair teams? Let's unlock our next set of code words!

MULTIPLICATION Clues	DIVISION Clues
• Times • Product of • Double (means × 2) • Triple (means × 3)	• Divided by • Shared equally • Split into groups • Half of (means ÷ 2)

Example: "A secret packet of crayons split into 3 boxes" becomes the beautiful math expression: $c \div 3$!

! Warning: The Tricky "Than" Trap!

Mathematicians have to be extra careful with one specific word trap. It is the word **"THAN"**!

When you see the phrase **"less than"**, it acts like a turnaround sign on a road. It tells you to swap the order of the items when you write down your expression!

Look at this example: *"4 less than a number x"*

- Many kids accidentally write: $4 - x$ (This is wrong!)
- The correct way is: $x - 4$ (This is right!)

Why? Think about it: if you have a pouch of marbles (x) and I take away 4, you start with the pouch first, then subtract the 4! Always put the number after "than" at the very front!

How to Solve Expressions (Plug it in!)

Because an expression has a secret hiding letter, you can't get a final number answer until someone hands you the magic key.

In math, solving an expression is called **Evaluating**. It's simple: your teacher will tell you what the secret letter equals, and you swap it out!

Problem Example: Evaluate the expression $y + 7$ if $y = 5$

- **Step 1:** Pop out the letter y.
- **Step 2:** Plug in the number 5 where y used to sit.
- **Step 3:** Add them together! $5 + 7 = 12$

You just solved the expression! It's like unlocking a secret level in a video game!

Real-World Practice: The Bakery Story

Let's put our skills to the test with a fun afternoon story problem!

Aisha is baking tasty cupcakes. She bakes an unknown number of vanilla cupcakes (v). Then she decides to bake 8 more chocolate cupcakes.

- **Task A: Write the expression for her total cupcakes.**
  We take her vanilla cupcakes (v) and add 8: $v + 8$
- **Task B: Solve it if we discover she baked $v = 10$ vanilla cupcakes!**
  We plug in 10: $10 + 8 = 18$ cupcakes in total!

Aisha made 18 delicious cupcakes! Problem solved!

Interactive Workbook Challenge!

Grab your favorite colored markers and let's translate these English phrases into clean math expressions:

1. The product of 5 and a number g:

- Clue: Product means multiplication!
- Your code expression: ___________________

2. 6 less than a number w:

- Clue: Watch out for the "than" turnaround trap!
- Your code expression: ___________________

Peek-a-boo Answers: *Challenge 1 is* $5 \times g$*. Challenge 2 is* $w - 6$*. You are fantastic!*

You are an Expression Wizard!

You have successfully mastered the secret art of writing and evaluating algebraic expressions! Let's lock in our 4 ultimate wizard rules:

Rule 1: Expressions have NO equal signs! They are just math phrases.

Rule 2: Spy Words tell you what sign to use (Sum = +, Difference = -).

Rule 3: Avoid the "Than" Trap! Flip the number order when you see "less than."

Rule 4: Evaluate means to plug the true number into the letter slot and hit compute.

CONGRATULATIONS! KEEP ROCKING MATH!

Wizard Expression Practice

Part A: Translate the words into math expressions! (Remember the "Than" Trap!)

1. The sum of a number x and 7.

Expression: ....

2. 5 less than a number y.

Expression: ....

3. The product of 4 and a number n.

Expression: ....

4. A secret number of cookies c divided equally among 2 friends.

Expression: ....

5. 9 increased by a number b.

Expression: ....

6. 3 taken away from a number h.

Expression: ....

Wizard Expression Practice (Continued)

Part B: Evaluate the expressions by plugging in the magic number key!

7. Evaluate the expression $k + 6$ if $k = 4$.

Your calculation & answer: ....

8. Evaluate the expression $12 - m$ if $m = 5$.

Your calculation & answer: ....

9. Evaluate the expression $3 \times z$ if $z = 6$.

Your calculation & answer: ....

10. Evaluate the expression $p \div 3$ if $p = 15$.

Your calculation & answer: ....

Part C: Real-World Wizard Challenges!

11. Ben has an unknown number of action figures a. He gives 4 to his brother. Write the math expression for the action figures Ben has left.

Expression: ....

12. Sara has t boxes of juice. Each box holds 6 juice pouches. Write the expression for her total juice pouches. Then, evaluate it if $t = 4$ boxes!

Expression: ....

If t = 4, Total Pouches = ...